



ori's fun/ repository

A Tour of Projects, Experiments & Ideas

20+

Projects

7

Categories

Python JS TS

Languages

2016 - 2026

Timespan

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■ Web Applications

Full-stack web apps built for real users — from ceramics studios to hotel management.

Inbal — Ceramics Studio Manager

vibe_coding/inbal/

A full-featured SaaS application for managing a ceramics studio. Students can view the weekly schedule, register for wheel/hand-building slots, manage a standby list, and handle transfers. Admins control the entire schedule.

- Student self-registration & standby queue
- Admin weekly/day calendar with drag-and-drop slot management
- Automatic standby notifications (WhatsApp planned)
- Authentication via Google / Facebook / Email
- Deployed on Vercel with Supabase PostgreSQL (Mumbai region)

Next.js 14

TypeScript

Prisma

Supabase

Vercel

Tailwind

```
Student      Admin      Database
■            ■            ■
■■■ View Schedule ■■■■■■ ■
■■■ Register Slot ■■■■■■ Validate ■■■■■■■■■■
■ (or join standby) ■■■■ Confirm ■■■■■■■■■■
■■■■ Confirmation ■■■■■■ ■
■            ■■■ Send Notification ■■■■
```

Road to Recovery — Ride Coordination

vibe_coding/road-to-recovery/

A ride coordination system for a non-profit rehabilitation association. Volunteers offer rides, patients request trips, and coordinators match them. Built with a focus on simplicity and accessibility.

- Patient ride requests with pickup / destination
- Volunteer availability and ride offer management
- Coordinator dashboard for matching and oversight
- Mobile-first responsive design

Next.js 14

TypeScript

Prisma

PostgreSQL

Tailwind

Uri's Hotel Management

[uri/hotel/](#)

A hotel management web application with rooms, guests, and reservations. Built as a monorepo with a React frontend and REST API backend.

- Room status tracking (available, occupied, maintenance)
- Guest profiles with search
- Reservation management dashboard
- Monorepo structure with shared packages

React

TypeScript

Vite

TanStack Query

Tailwind

Oti's ChatGPT Web App

[oti/chatgpt_api_example/](#)

A Flask web application that wraps the OpenAI API with a simple chat interface. Built on OpenAI's quickstart example, extended with session history and personality prompts.

- Stateful conversation via Flask sessions
- Personality and summarization prompt helpers
- Clean HTML/CSS frontend

Python

Flask

OpenAI API

Pipin — Middle Earth Adventure Game

[vibe_coding/pipin/](#)

A browser-based text adventure set in Tolkien's Middle Earth. The player controls Pippin (Peregrin Took) in a world faithful to the lore. A dynamic narrative engine powered by Gemini generates story responses. Every location has a unique AI-generated illustration.

- ~30 hand-placed locations with individual artwork
- Dynamic narrative via Gemini 1.5 Pro — story diverges from the books
- Hebrew RTL interface with a modern, bright design
- AI image generation for locations and NPC portraits
- World graph: locations, connections, items, characters

HTML/CSS/JS

Gemini API

Nano Banana

RTL

Player Action

```

  ▼
game.js ■■■ world.js (locations / NPCs / items)
  ▼
prompts.js ■■■ api.js ■■■ Gemini 1.5 Pro
                ■
                ■■■■■■■■■■ story response + choices ■■■■■■■■■■
```

■ Machine Learning & AI

Deep learning models, transformers, GANs, fine-tuning, and LLM-powered tools.

Transformer from Scratch

[ml/transformer/](#)

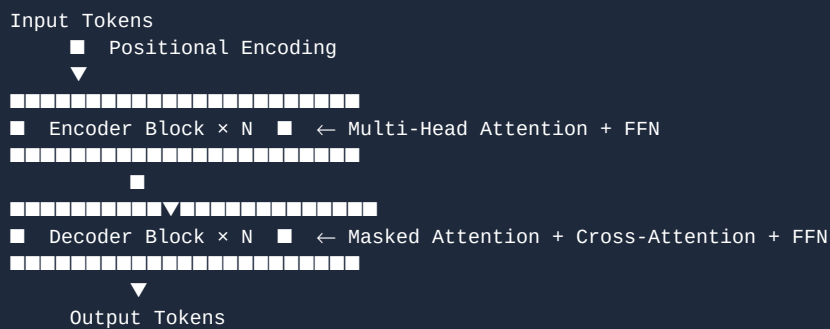
A clean, educational implementation of the original Transformer architecture ('Attention Is All You Need', Vaswani et al. 2017) in PyTorch. Built alongside Aladdin Persson's YouTube tutorial series.

- Multi-head self-attention with scaled dot-product
- Encoder / Decoder blocks with residual connections
- Positional encoding, feed-forward layers
- Mask support for decoder (causal masking)

Python

PyTorch

Transformer



Llama 2 Fine-tuning

[ofer/myenv/uri_llama2_finetune.py](#)

Fine-tuning Meta's Llama 2 language model using QLoRA (quantized LoRA), PEFT, and the TRL library. Includes a Gradio interface for streaming inference. Built with Ofer for a specific instruction-following task.

- 4-bit quantization via BitsAndBytes (QLoRA)
- LoRA adapters via PEFT — efficient parameter tuning
- SFTTrainer from TRL for supervised fine-tuning
- W&B; integration for experiment tracking
- Streaming output via Gradio UI

Python

PyTorch

Llama 2

PEFT

QLoRA

Gradio

W&B

MNIST Classifiers

tensorflow/

Two implementations of the classic handwritten digit classifier on MNIST: a simple softmax model and a convolutional network (CNN) — both written in TensorFlow 1.x as learning exercises.

- `mnist.py` — softmax regression baseline
- `mnist_conv.py` — 2-layer CNN with max pooling
- Weight/bias initialization helpers
- Interactive TensorFlow session (TF1 style)

Python

TensorFlow 1

CNN

MNIST

■ Reinforcement Learning

Agents learning to play games through interaction — from 2048 to CartPole.

2048 — RL Agent Suite

[rl/2048/gym2048o/](#)

A full reinforcement learning environment for the game 2048, packaged as an OpenAI Gym-compatible environment. Multiple agents were developed and compared.

- Custom Gym environment with proper reset/step/render API
- Random Agent — uniform random baseline
- Cross Entropy Agent — policy gradient with experience replay
- DQN Agent — Deep Q-Network with epsilon-greedy exploration
- Search Agent — tree search heuristic
- Tkinter GUI for human play, keyboard UI for testing
- TensorBoard integration for training metrics

Python

PyTorch

OpenAI Gym

DQN

Cross Entropy

Tkinter

```
Agent                                Environment (2048 Grid)
  ■                                ■
  ■■■ action (0-3: U/D/L/R) ■■■
  ■■■ state, reward, done ■■■
  ■                                ■
Neural Net                          ■
Q(s,a) ■■■ argmax ■■■■■■■■■■
  ■
Experience Replay Buffer
```

CartPole — Cross Entropy Agent

[rl/2048/gym2048o/cart_pole_cea.py](#)

A cross entropy method agent applied to OpenAI Gym's CartPole environment — a simpler testbed used to validate the agent before applying it to 2048.

- Policy network trained on elite episodes
- Validates cross entropy method on a well-understood benchmark

Python

PyTorch

OpenAI Gym

Cross Entropy Method

■ AI Tools & Utilities

Practical tools that integrate AI APIs into everyday workflows.

Voice-to-Claude

vibe_coding/voice_to_claude.py

A hands-free voice input system for Claude Code. Listens continuously using a microphone, detects speech automatically with VAD (Voice Activity Detection), transcribes with Whisper, and pastes directly into the terminal.

- Continuous listening — no button press needed
- Voice Activity Detection: starts/stops on speech automatically
- Whisper transcription (local, small model)
- Hebrew language support (language='he')
- Auto-pastes into active terminal via xdotool

Python

Whisper

VAD

pyaudio

Hebrew

```
Microphone
  ■
  ▼
VAD (Voice Activity Detection)
  ■ speech detected
  ▼
Record audio buffer
  ■
  ▼
Whisper (local) ■■■ Hebrew text
  ■
  ▼
xdotool type ■■■ paste into Claude Code terminal
```

WhatsApp Voice Transcriber

vibe_coding/openclaw/whatsapp_trans/

A transcription tool for WhatsApp voice messages, integrated as an OpenClaw provider. Receives OGG/Opus audio files, converts to WAV using ffmpeg, and transcribes with Google Cloud Speech-to-Text.

- OGG/Opus → 16kHz mono WAV conversion via ffmpeg
- Google Cloud STT for accurate Hebrew/multilingual transcription
- OpenClaw CLI provider interface
- Voice agent wrapper for full pipeline automation

Python

Google Cloud STT

ffmpeg

OpenClaw

WhatsApp

Wikipedia Information Extractor

ie/

A set of scripts for extracting structured information from Wikipedia via the MediaWiki API. Progressively more complex queries: categories, revisions, page content extraction.

- query1–4.py: incrementally complex Wikipedia API queries
- Category listing with prefix filtering
- Page revision and content retrieval

Python

Wikipedia API

urllib

■ Math & Science Notebooks

Interactive explorations of mathematics, physics, and algorithms.

Infinitesimal Calculus 2 — Technion

[colabs/inf2.ipynb](#)

Interactive Jupyter notebook following the Technion's Calculus 2M course by Dr. Aviv Tsenzor. Covers topics from sequences and series to multi-variable calculus, with interactive visualizations.

- Sequences, limits, and series convergence
- Multi-variable functions and partial derivatives
- Interactive Plotly/Matplotlib visualizations
- Hebrew + English mixed notes

Python

Jupyter

NumPy

Matplotlib

Plotly

Function Fitter

[colabs/FunctionFitter.ipynb](#)

A notebook for fitting mathematical functions to data points. Explores curve fitting methods and visualization of residuals.

- Polynomial and non-linear curve fitting
- Residual analysis and goodness-of-fit metrics

Python

Jupyter

SciPy

NumPy

Matplotlib

ESLA — Physics & Math Simulations

[esla/](#)

A collection of Python simulations from an early period — physical simulations, 3D visualizations, and mathematical tools. Originally managed in SVN (pre-Git era).

- 3dFunction.py — 3D surface plotter
- vec_comb.py — vector combination visualizer (VPython)
- rectInt.py — rectangular integration approximation
- gEliminate.py — Gaussian elimination solver
- erez_ir.py — city/country data processor
- primes/ — Sieve of Eratosthenes, Ulam spiral, double primes
- surf/ — surface rendering experiments

Python

VPython

Matplotlib

NumPy

tkinter

■ Games & Interactive

Games, puzzles, and interactive visual projects.

Sudoku Solver

sudoku/sudoku.py

A Sudoku puzzle solver and interactive board built with Python's tkinter. Supports both manual puzzle entry and automated solving.

- Interactive tkinter canvas UI
- Constraint-based solver algorithm
- Edit mode for entering custom puzzles

Python

tkinter

Scratch Projects

scratch/

A collection of visual programming projects from Scratch — an early exploration of algorithms and interactive graphics.

- Koch Snowflake — recursive fractal
- Sierpinski Triangle — recursive fractal
- Tower of Hanoi — animated auto-solution
- Find Path in Maze — pathfinding visualization
- Breakout clone
- Big World — scrolling and zooming camera
- Bat Shooter, Bouncing Ball, Obstacle Race

Scratch

Algorithms

Fractals

Pathfinding

Breakout (Python)

misc/breakout.py

A Python implementation of the classic Breakout arcade game, demonstrating event-driven programming and basic game loop concepts.

- Paddle, ball, and brick collision detection
- Score tracking and game-over conditions

Python

tkinter / pygame

Empires — History Timeline

vibe_coding/ori/ and animate/

An interactive browser-based visualization of world empires throughout history. Shows the rise and fall of civilizations on a timeline with animated transitions. Available in Hebrew, English, and Spanish.

- SVG/Canvas-based timeline animation
- Multilingual: Hebrew (RTL), English, Spanish
- Mobile-responsive layout
- Search and filter by civilization

JavaScript

HTML/CSS

SVG/Canvas

RTL

■ Classic & Legacy

Older experiments and learning projects that show the journey.

CheckProj — Django Starter

[checkProj/](#)

A minimal Django project used to explore Python web development. Includes models, views, URL routing, and template rendering.

- Django MVC structure (models, views, templates)
- Basic URL routing

Python

Django

Python Features Notebook

[python_features/python_features.ipynb](#)

A reference notebook exploring Python language features — decorators, generators, comprehensions, and more.

- Language feature demonstrations with examples
- Used as a personal reference / cheatsheet

Python

Jupyter

Misc Utilities

[misc/](#)

A grab-bag of standalone Python scripts exploring algorithms and data structures.

- `priorotyQueue.py` — custom priority queue implementation
- `gcd.py` — greatest common divisor variants
- `shablulPrimes.py` — prime number experiments
- `writeltself.py` — a quine (program that prints its own source)

Python

Algorithms

Data Structures

C++ / OpenGL

[cpp/opengl/](#)

Early experiments with C++ and OpenGL for 3D graphics rendering.

- OpenGL rendering pipeline
- 3D geometry and transformations

C++

OpenGL

■ Summary

Category	# Projects	Primary Tech
Web Applications	5	Next.js, React, TypeScript
Machine Learning & AI	6	PyTorch, TensorFlow, LLMs
Reinforcement Learning	2	PyTorch, OpenAI Gym
AI Tools & Utilities	3	Python, Whisper, Cloud APIs
Math & Science	3	Python, Jupyter, NumPy
Games & Interactive	4	Python, JavaScript, Scratch
Classic & Legacy	4	Python, Django, C++

This repository spans a decade of experimentation — from early Python simulations and Scratch games to production-grade Next.js applications and LLM fine-tuning. Each project reflects a different curiosity, collaboration, or real-world need.